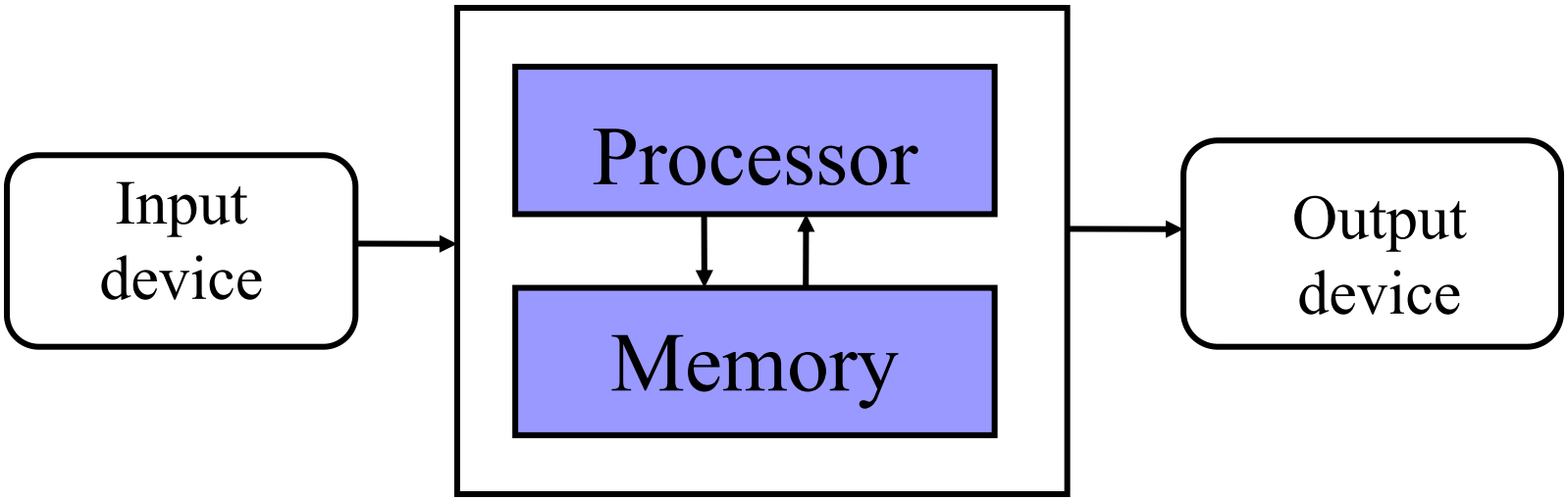
Main Memory & Memory Hierarchy

Outline

- The von Neumann Model (reminder)
- Memory Address
- RAM
- ROM
- Memory Hierarchy

The von Neumann Model (reminder)



- In the Von Neumann architecture, main memory serves as the exclusive workspace for the CPU.

The von Neumann Model (reminder)

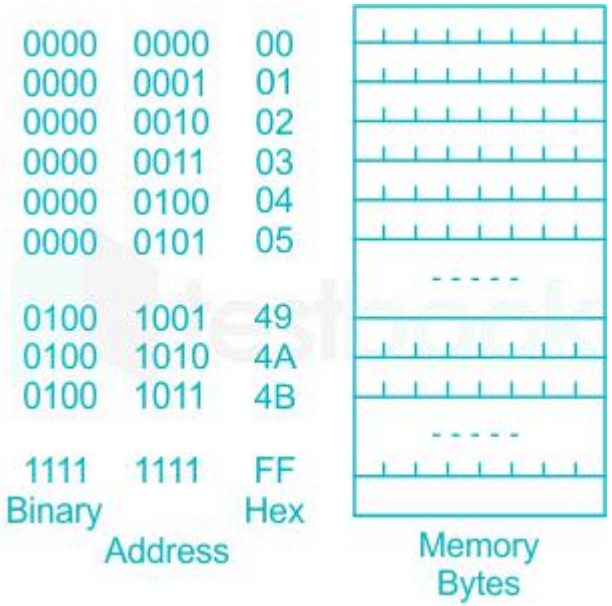
- The CPU fetches instructions and data exclusively from main memory, making it the critical.
- Input devices transmit external information into this memory space, while output devices read from it to present results to the user.
- The total volume of main memory determines the maximum number of processes and the addressable workspace.

Main memory

- Two important properties of main memory
  - Physical memory internal to the computer.
  - CPU can manipulate only data that is inside the main memory.

Memory is byte-addressable

- The smallest unit of memory is a bit.
- Accessing a single bit at a time isn't very efficient
- Accessesing multiple grids of 1-bit memory cells in parallel is more efficient

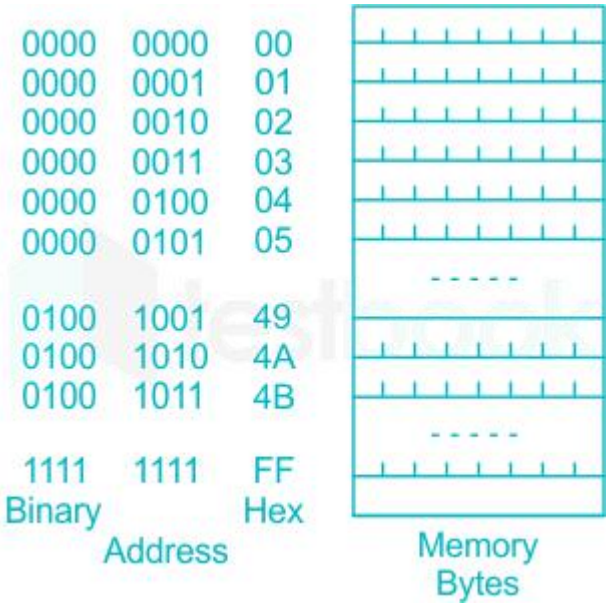


SHL shl eax, 3

SHR shr eax, 1

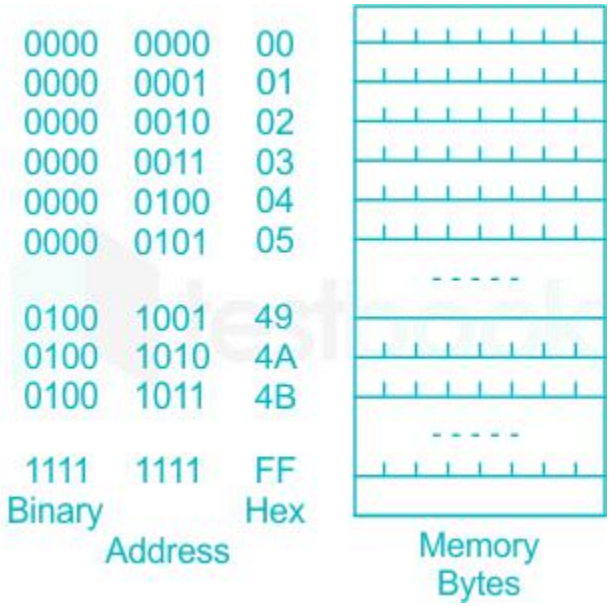
Addressing in Memory

- Memory address: a numeric value that identifies the location of a set of bits in memory
- Memory is typically byte-addressable, meaning a single memory address refers to 8 bits of data



Main memory: RAM

- Today's main memory is tyically Random Access Memory (RAM).
  - In old days with magnetic tape: accessing subsequent data requires traversing preceding data.
  - In RAM, you can access data stored in the second or following blocks without going through the first block of data.

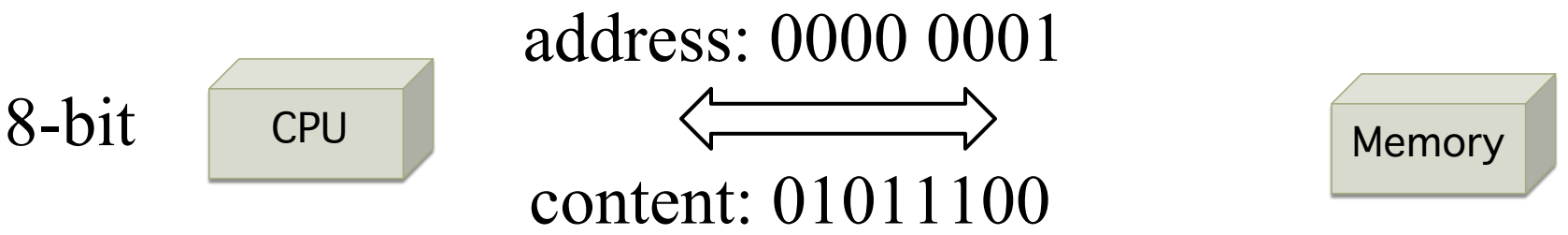


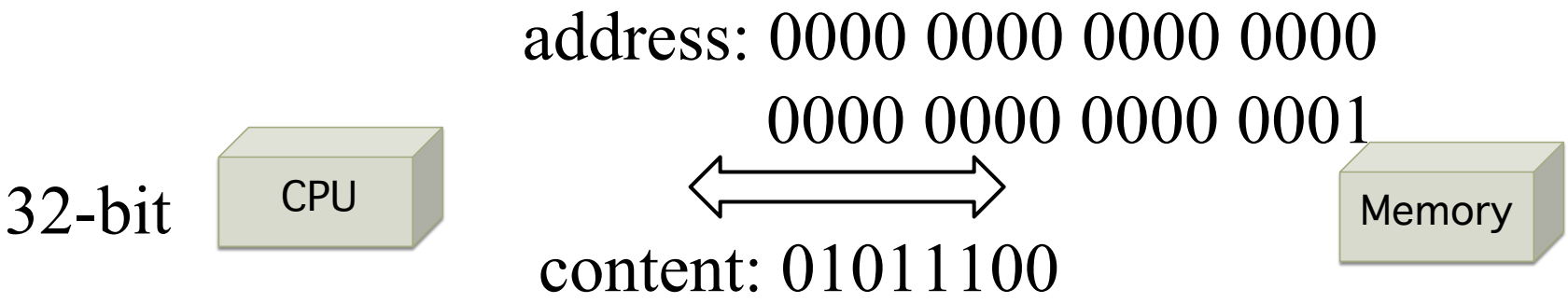
Address width

- Address width is determined by:
  - The number of bits in the CPU address registers.
  - The number of lines in the address bus.
  - E.g,
    - 64-bit CPU;
    - 32-bit CPU;
    - 8-bit CPU

Address width (cont.)

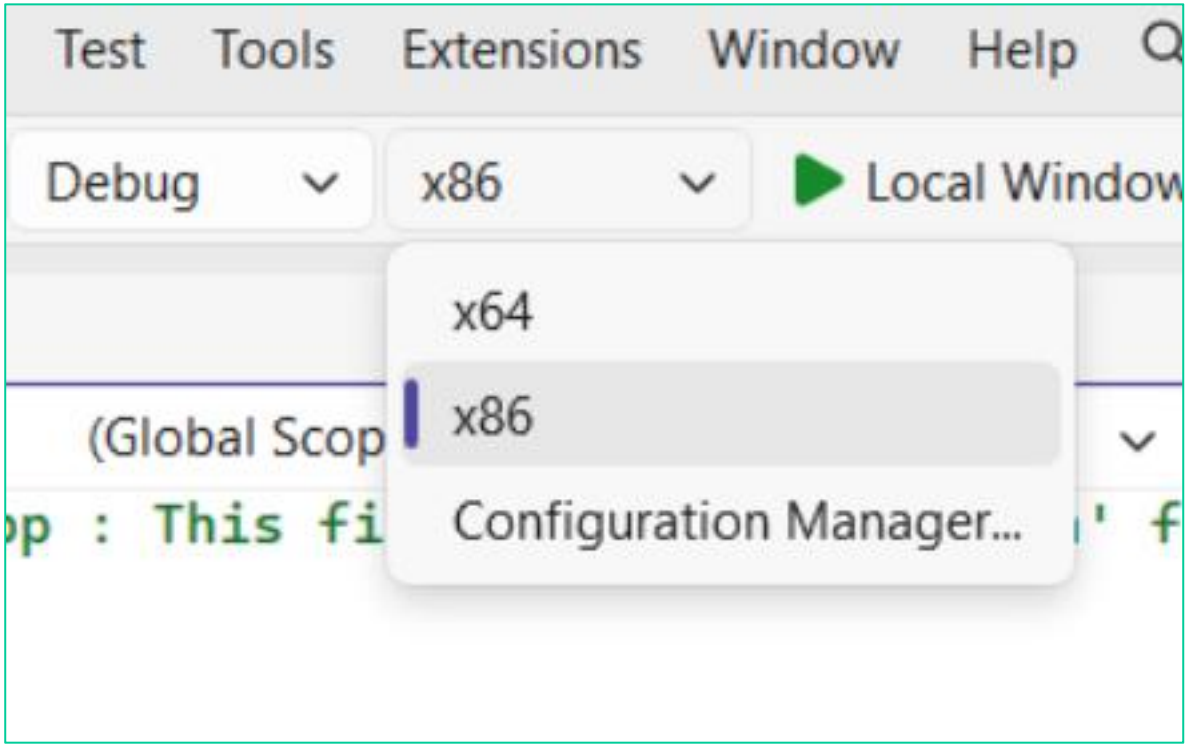
- Example: 8-bit and 32-bit CPU





Address width (cont.)

- Recall this in our lab:



Address width and memory length

Address width			Maximal memory length	
16			**64**	**Kbytes**
32	**Pentium**		**4**	**Gbytes**
64	**64-bit architectures**	**>**	**17 billion Gbytes**	

Modern computing systems have 64-bit addressing

- A 64-bit address range should suffice for a long time as far as memory is concerned.
  - Unlikely you will ever put 2^64 bytes of memory into a computer system and feel that you need more.
- Till the appearance of large language model training, which could use memory a bus of 4,096 bits or even more.

Addresses in Assembly

- Can you tell the difference ?
  - LEA
  - MOV

```
...
int myarray[5]; // assume an address of 0x1234 for myarray
myarray[0] = 2;
myarray[1] = 4;
myarray[2] = 6;
myarray[3] = 8;
myarray[4] = 10;
_asm{
    MOV EAX, myarray
    LEA   EBX, myarray
}
```

Instruction Formats

- Byte ordering, or endianness, is a major architectural consideration.
- If we have a two-byte integer, the integer may be stored in wo different ways.
  - In little endian machines, the least significant byte is followed by the most significant byte.
  - Big endian machines store the most significant byte first (at the lower address).

Instruction Formats (cont.)

- For example, suppose we have the hexadecimal number 12345678.
- The big endian and small endian arrangements of the bytes are shown below.

Address →	00	01	10	11
Big Endian	12	34	56	78
Little Endian	78	56	34	12

Instruction Formats (cont.)

- Big endian: more natural.
  - The sign of the number can be determined by looking at the byte at address offset 0.
  - Strings and integers are stored in the same order.
- Little endian:
  - Easier to convert in different width.
  - Conversion from a 32-bit integer address to a 64-bit integer address does not require any arithmetic.

Addresses in Assembly (cont.)

- Can you tell the difference between a *value* and a [*value*] ?

```
...
int myarray[5]; // assume an address of 0x1234 for myarray
myarray[0] = 2;
myarray[1] = 4;
myarray[2] = 6;
myarray[3] = 8;
myarray[4] = 10;
_asm{
    MOV  EBX, 0x1234
    MOV  EAX, [0x1234]
}
```

Question

- What is the value in EBX?

```
...
int myarray[5]; //
myarray[0] = 2;
myarray[1] = 4;
myarray[2] = 6;
myarray[3] = 8;
myarray[4] = 10;
_asm{
    LEA EAX, myarray
    MOV  EBX, [EAX+4]
}
```

```
...
char myarray[3]; //
myarray[0] = 'a';
myarray[1] = 'b';
myarray[2] = 'c';
myarray[3] = 'd';
myarray[4] = 'e';
_asm{
    LEA EAX, myarray
    MOV  EBX, [EAX+4]
}
```

Words, Bytes and Bits



- A standard unit of 8 bits is called a **byte**.
- A word consists of one of multiple **bytes**. Its size reflects the CPU's natural data-processing width.
  - A word consists of 4 bytes in x86 (32 bit) CPU
  - A word consists of 8 bytes in 64-bit CPU

Kilobytes, Megabytes, etc

- 1 Byte (B) = 8 bits.
- 1 Kilobyte (KB) = 2^{10} bytes = 1024 bytes.
- 1 Megabyte (MB) = 2^{10} KB = 1024 KB.
- 1 Gigabyte (GB)= 2^{10} MB = 1024 MB.
- 1 Terabyte (TB) = 2^{10} GB = 1024 GB.

Exercise. Compute how many bytes in one Megabyte and in one Gigabyte.

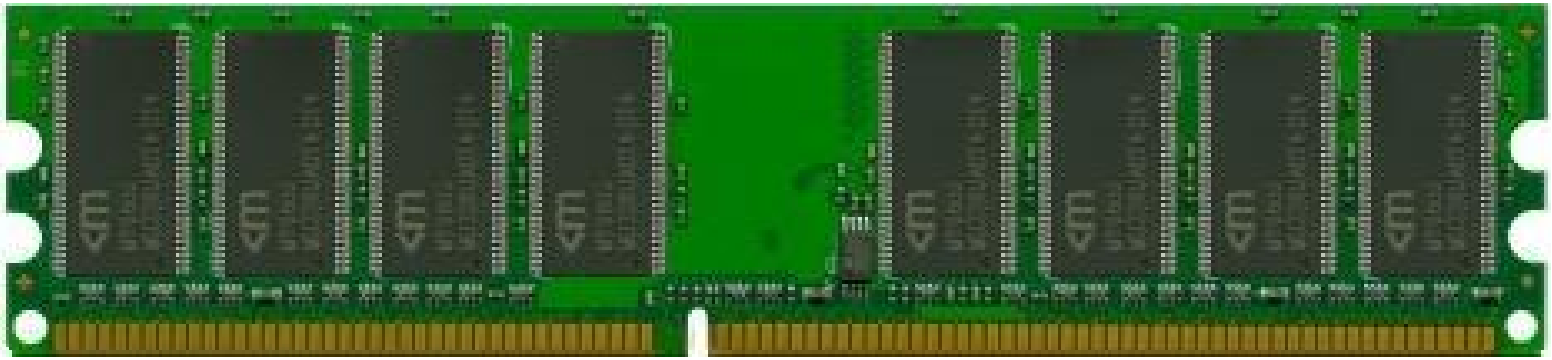
Typical sizes

- Main memory capacity is strictly measured using binary prefixes based on powers of 1024 (2^10), reflecting the binary nature of digital addressing.
  - Typical RAM module has 1/2/4/8 GB of memory.
  - Common size of CD is 650 MB.
  - Common size of memory disc is 1+ GB.
  - Typical hard drive has the size of hundreds to thousands GB.

Real-world Data Scale

- The complete literary works of William Shakespeare require approximately 8 MB of storage, illustrating the extreme compactness of pure text data. This includes 38 plays, 154 sonnets, and numerous poems.
- The human genome contains roughly 3 billion base pairs, translating to approximately 0.8 GB of raw data — demonstrating the remarkable efficiency of biological encoding across 23 chromosome pairs.
- Today's large language model may have a size of hundreds of GB

Memory module



Back to RAM

- There are two basic types of RAM:
  - Dynamic RAM (DRAM).
    - Cheaper, but slower.
    - Implemented via capacitors.
    - DRAM needs to be ***refreshed.***
  - Static RAM (SRAM).
    - Faster, but more expensive.
    - Implemented via flip-flops.
    - No need for refreshing.
- Both types of RAM are **volatile**.
  - They lose their contents when the power is turned off.
  - Both requires continuous electrical power to maintain stored states.

DRAM Basics

- DRAM stores each bit as charge in a microscopic capacitor paired with a transistor.
  - Simple architecture: Each bit is stored as the presence or absence of electrical charge in **a capacitor** controlled by **a transistor**.
  - Extreme density: microscopic capacitors enable massive storage density, with **billions of bits per chip**
  - Charge leakage: stored data degrades in milliseconds due to parasitic resistance, requiring continuous refresh by the memory controller.

DRAM Refresh

- DRAM refresh is a necessary to counteract capacitor leakage.
  - Typically occurring every 64 milliseconds
  - The memory controller must sequentially read and rewrite every row in the array.

SRAM

- SRAM achieves its exceptional speed and stability by utilizing a 6-transistor flip-flop circuit to store each bit.
  - high stability and near-instantaneous access times
  - 6-transistor cell's large silicon footprint yields low density and high cost

Discussion

- DRAM used as the main memory.
- Where is SRAM used?

DRAM Performance

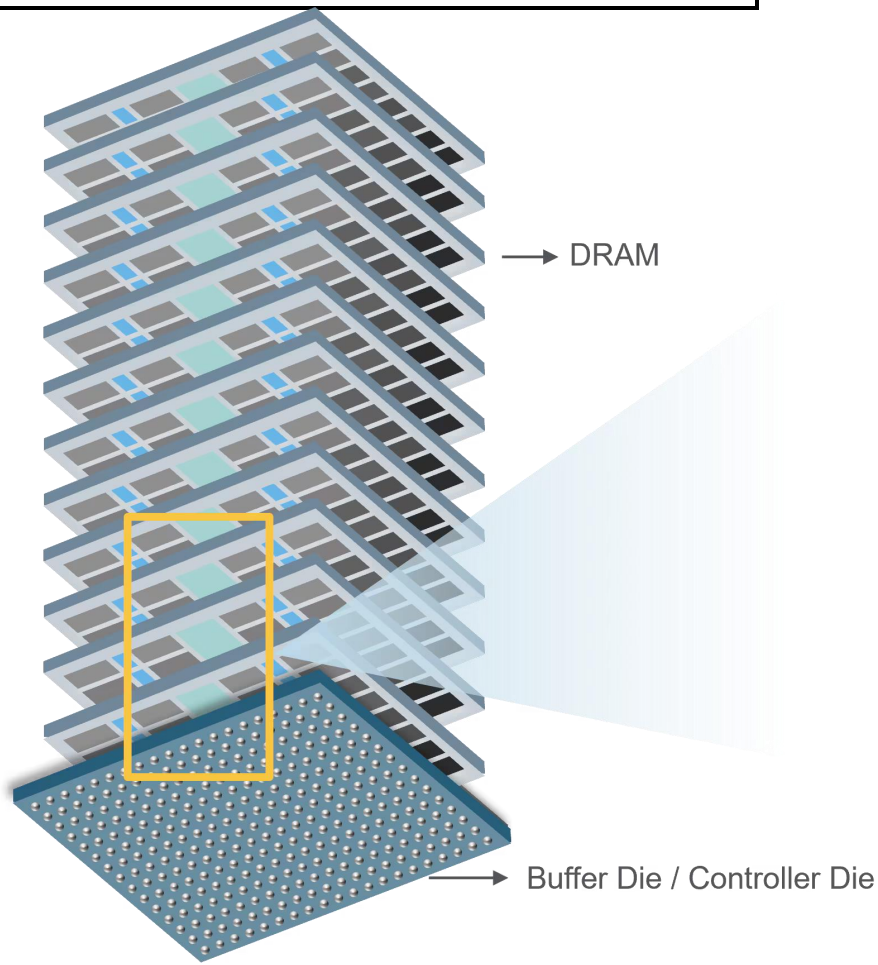
- Two key metrics: latency and bandwidth
- Latency
  - Measured in nanoseconds, latency captures the time delay between a memory request and the arrival of the first data word. .
- Bandwidth
  - Measured in Gigabytes per second, bandwidth captures the aggregate volume of data transferred per second.

Emerging Memory Tech: GDDR

- The exponential growth of AI has pushed forward high-bandwidth memory (HBM).
- GDDR: Graphics Double Data Rate
  - Sacrifices latency and density to achieve extremely high clock frequencies and wide bus widths

Emerging Memory Tech: HBM

- HBM 3D Stacking:
  - Stacks multiple RAM dies vertically to create a 3D structure that shortens signal paths and reduces power consumption.



HBM Stack

From RAM to ROM

- Read-only memory (ROM)
- The volatility of RAM necessitates write operations to non-volatile storage, ensuring data persistence across power cycles.
- ROM is non-volatile.

ROM Basics

- Early days
  - permanent, factory-programmed silicon
- Now
  - highly flexible, electrically erasable Flash memory



ROM Types

- Mask ROM
  - Programmed during semiconductor fabrication and cannot be altered.
- EPROM
  - Data can be erased via ultraviolet light exposure through a quartz window on the chip package.
- EEPROM
  - enabling electrical erasure of the data
  - Flash memory is technically a special type of EEPROM

Firmware and BIOS

- Firmware
  - programming designed to help hardware devices
  - also known as ‘software for hardware’
- BIOS - Basic Input Output System
- ROM to store the BIOS/UEFI firmware
  - CPU needs a hardcoded starting address when the computer is power-up.
  - main memory is volatile and empty at power-up.

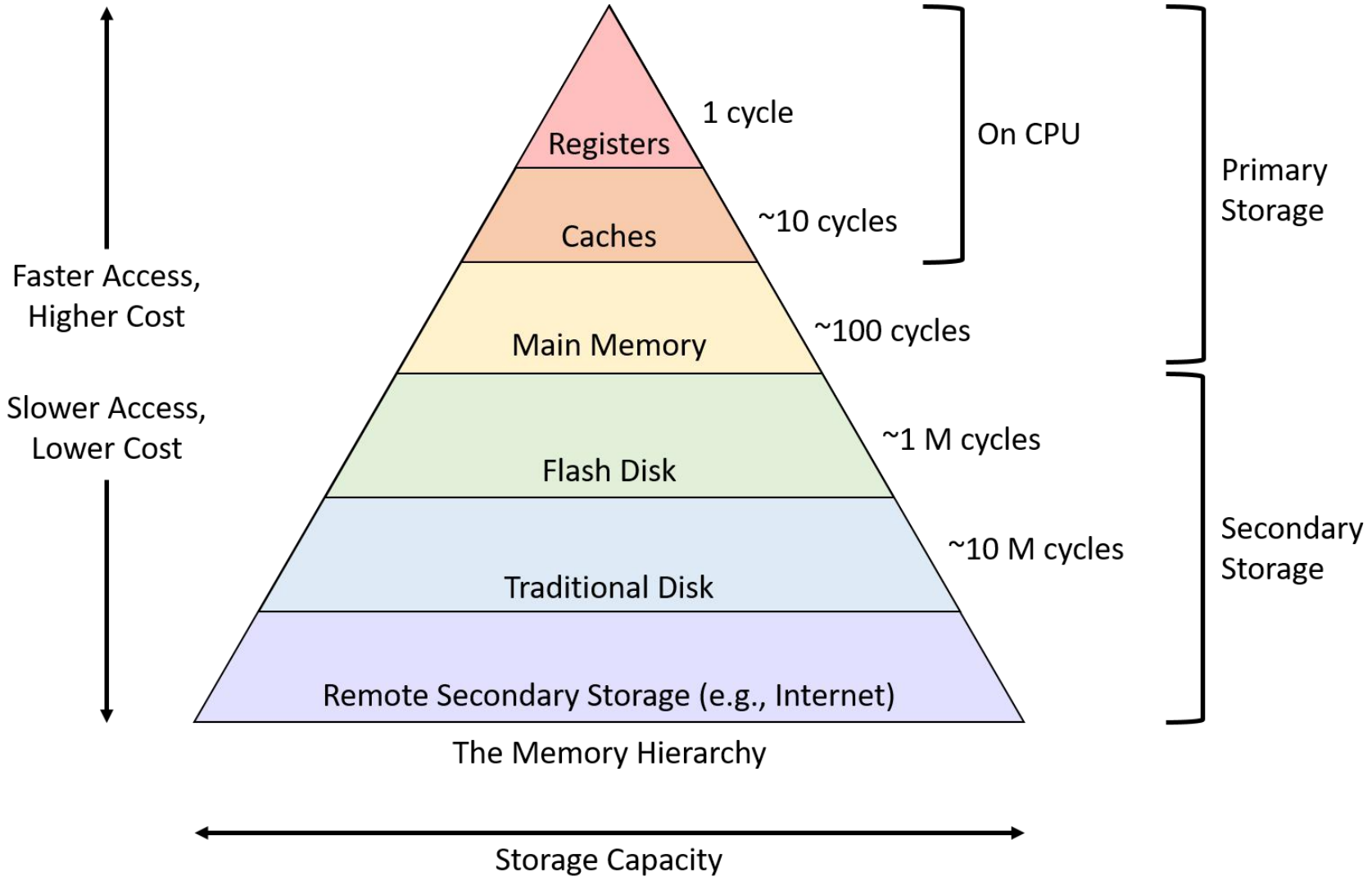
Memory Hierarchy

- Memory lies at the heart of the stored-program computer.
- We studied the ways in RAM is accessed by assembly codes.
- How about the other part of the memroy

Registers

- Registers are the memory cells which are core part of the processor itself.
- It has very fast access (a few nanoseconds).
- Not that much memory: tens of 32-, 64-, 80-bit registers (typically).

Memory Pyramid



Memory Pyramid

- Going down the hierarchy:
  - Increased capacity.
  - Increased access time.
  - Decreased frequency of access of the memory by the processor.
  - Decreased cost per bit.

Why Hierarchy

- Goal: to provide the best performance at the lowest cost.
  - Faster memory is more expensive than slower memory.
  - Small, fast storage elements are kept in the CPU, larger, slower main memory is accessed through the data bus.
  - Larger, (almost) permanent storage in the form of disk drives is still further from the CPU.

Data Flow in Memory Hierarchy

- To access a particular piece of data, the CPU first sends a request to its nearest memory, usually cache.
- If the data is not in cache, then main memory is queried. If the data is not in main memory, then the request goes to disk.
- Once the data is located, then the data, and a number of its nearby data elements are fetched into cache memory.

Q&A

- Q. Which of the following has fastest access?
- A. register
- B. cache
- C. main memory
- D. disk

*

Q&A

- Q. which of the following is most expensive in terms of cost?
- A. register
- B. cache
- C. disk
- D. tape

*

Q&A

- Suppose we have a small computer system in which the bus width is 18. What is the maximum memory space this computer system can have full access to